

Section 1 - 125742/0/0

0001 --> 125742/0.0 (Original Application) - Recd 2021-05-06 - DATS# 1067058
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[0001] Transfer of Obligation
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4.2	Study Reports
4.2.1	Pharmacology
4.2.1.1	Primary Pharmacodynamics
[0001]	R-20-0085 - COVID-19: Immunogenicity Study of the LNP-Formulated modRNA Encoding the Viral S Protein-V3
[0001]	R-20-0112 - Characterizing the Immunophenotype in Spleen and Lymph Node of Mice Treated with SARS-CoV-2 Vaccine Candidates
[0001]	R-20-0211 - In vitro Expression of BNT162b2 Drug Substance and Drug Product
[0001]	VR-VTR-10671 - BNT162b2 (V9) Immunogenicity and Evaluation of Protection against SARS-CoV-2 Challenge in Rhesus Macaques
[0001]	VR-VTR-10741 - Structural and Biophysical Characterization of SARS-CoV-2 Spike Glycoprotein (P2 S) as a Vaccine Antigen
4.2.2	Pharmacokinetics
4.2.2.2	Absorption
[0001]	072424 - A Single Dose Pharmacokinetics Study of ALC-0315 and ALC-0159 Following Intravenous Bolus Injection of PF-07302048 Nanoparticle Formulation in Wistar Han Rats
4.2.2.3	Distribution
[0001]	R-20-0072 - Expression of Luciferase-Encoding ModRNA After I.M. Application of GMP-Ready (b)(4) NP Formulation
[0001]	185350 - A Tissue Distribution Study of a [3H]-Labelled Lipid Nanoparticle-mRNA Formulation Containing ALC-0315 and ALC-0159 Following Intramuscular Administration in Wistar Han Rats
4.2.2.4	Metabolism
[0001]	01049-20008 - In Vitro Metabolic Stability of ALC-0315 in CD-1/ICR Mouse, Sprague Dawley Rat, Wistar Han Rat, Cynomolgus Monkey, and Human Liver Microsomes
[0001]	01049-20009 - In Vitro Metabolic Stability of ALC-0315 in CD-1/ICR Mouse, Sprague Dawley Rat, Cynomolgus Monkey, and Human Liver S9 Fractions
[0001]	01049-20010 - In Vitro Metabolic Stability of ALC-0315 in CD-1/ICR Mouse, Sprague Dawley Rat, Wistar Han Rat, Cynomolgus Monkey, and Human Hepatocytes
[0001]	01049-20020 - In Vitro Metabolic Stability of ALC-0159 in CD-1/ICR Mouse, Sprague Dawley Rat, Wistar Han Rat, Cynomolgus Monkey, and Human Liver Microsomes
[0001]	01049-20021 - In Vitro Metabolic Stability of ALC-0159 in CD-1/ICR Mouse, Sprague Dawley Rat, Cynomolgus Monkey, and Human Liver S9 Fractions
[0001]	01049-20022 - In Vitro Metabolic Stability of ALC-0159 in CD-1/ICR Mouse, Sprague Dawley Rat, Wistar Han Rat, Cynomolgus Monkey, and Human Hepatocytes
[0001]	043725 - Investigation of the Biotransformation of ALC-0159 and ALC-0315 In Vitro and In Vivo in Rats
4.2.3	Toxicology
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[0001]	38166 - Repeat-Dose Toxicity Study of Three LNP-Formulated RNA Platforms Encoding for Viral Proteins By Repeated Intramuscular Administration to Wistar Han Rats
[0001]	20GR142 - 17-Day Intramuscular Toxicity Study of BNT162b2 (V9) and BNT162b3c in Wistar Han Rats with a 3-Week Recovery
4.2.3.5	Reproductive and Developmental Toxicity
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[0001]	20256424 - Combined Fertility and Developmental Study (Including Teratogenicity and Postnatal Investigations) of BNT162b1, BNT162b2 and BNT162b3 by the Intramuscular Route in the Wistar Rat GLP Study
4.3	Literature References
4.3	Literature References
[0001]	Al-Amri SS, Abbas AT, Siddiq LA, et al. Immunogenicity of candidate MERS-CoV DNA vaccines based on the spike protein. Sci Rep 2017;7:44875
[0001]	Bany MA, May S, Wesver EA. Imaging luciferase-expressing viruses. Methods Mol Biol 2012; 797:79-87.
[0001]	Baum A, Fulton BO, Wioga E, et al. Antibody cocktail to SARS-CoV-2 spike protein prevents rapid mutational escape seen with individual antibodies. Science 2020; :First release
[0001]	Berger Rentsch M, Zimmer G. A vesicular stomatitis virus replicon-based bioassay for the rapid and sensitive determination of multi-species type I interferon. PLoS One 2011;6(10):e25858
[0001]	Boone L, Meyer D, Cusick P, et al. Selection and interpretation of clinical pathology indicators of hepatic injury in preclinical studies. Vet Clin Pathol. 2005;34(3):182-188.
[0001]	Brouwer PJM, Canale TG, van der Straten K, et al. Potent neutralizing antibodies from COVID-19 patients define multiple targets of vulnerability. Science 2020;359(6504):843-850.
[0001]	Cal Y, Zhang J, Kiao T, et al. Distinct conformational states of SARS-CoV-2 spike protein. Science 2020;10.1126/science.abd4251
[0001]	Chen C-Y, Tran DM, Cavedon A, et al. Treatment of Hemophilia A Using Factor VIII Messenger RNA Lipid Nanoparticles. Mol Ther Nucleic Acids. 2020;20:534-44.
[0001]	de Wit E, N van Doremalen, D Faizaran, et al. SARS and MERS: recent insights into emerging coronaviruses. Nat Rev Microbiol 2016;14(8):523-34
[0001]	Elie U, Ramishetti S, Dammes N, et al. Design of SARS-CoV-2 RBD mRNA Vaccine using Novel Ionizable Lipids. bioRxiv 2020.10.1101/2020.10.15.341537.
[0001]	Ennulat D, Magid-Slav M, Rehm S, et al. Diagnostic performance of traditional hepatobiliary biomarkers of drug-induced liver injury in the rat. Toxicol Sci 2010;116(2)(Aug):397-412.
[0001]	Fukuchi M, Saito R, Maki S, et al. Visualization of activity-regulated BDNF expression in the living mouse brain using non-invasive near-infrared bioluminescence imaging. Mol Brain 2020. 13(1):122.
[0001]	Hassett KJ, Benenato KE, Jacquinet E et al. Optimization of lipid nanoparticles for intramuscular administration of mRNA vaccines. Mol Ther Nucleic acids 2019;15:1-11.
[0001]	Henderson R, Edwards RJ, Mansouri K, et al. Controlling the SARS-CoV-2 spike glycoprotein conformation. Nat Struct Mol Biol 2020;(Jul):Online
[0001]	Hoffmann M, Kleine-Weber H, Schroeder S, et al. SARS-CoV-2 cell entry depends on ACE2 and TMPRSS2 and is blocked by a clinically proven protease inhibitor. Cell 2020. 181(2):271-280.e8
[0001]	Hultwitt RJ, de Haan CA, Bosch BJ. Coronavirus spike protein and tropism changes. Adv Virus Res 2016;96:25-57
[0001]	Hutloff A. Regulation of T follicular helper cells by ICOS. Oncotarget 2015;6(26)(Sep):21785-6.
[0001]	Jeon YH, Choi Y, Kang JH, et al. In vivo monitoring of DNA vaccine gene expression using firefly luciferase as a naked DNA. Vaccine 2006; 24(16):3067-62.
[0001]	Jiang S, Hyllyer C, Du L. Neutralizing antibodies against SARS-CoV-2 and other human coronaviruses. Science and society. Trends Immunol 2020; 41(5)(May):355-9
[0001]	Karko K, Buckstein M, Ni H, et al. Suppression of RNA recognition by Toll-like receptors: the impact of nucleoside modification and the evolutionary origin of RNA. Immunity 2005;2(Aug):165-75.
[0001]	Kaushal D, Foreman TW, Gautam US, et al. Mucosal vaccination with attenuated Mycobacterium tuberculosis induces strong central memory responses and protects against tuberculosis. Nat Commun 2015; 6:8533.

[0001] Kaushal D, Foreman TW, Gautam US, et al. Mucosal vaccination with attenuated *Mycobacterium tuberculosis* induces strong central memory responses and protects against tuberculosis. *Nat Commun* 2015; 6:8533.

[0002] Ke Z, Oton J, Qu K, et al. Structures and distributions of SARS-CoV-2 spike proteins on intact virions. *Nature* 2020;10.1038/s41586-020-2665-2.

[0003] Kim JY, Ko JH, Kim Y, et al. Viral load kinetics of SARS-CoV-2 infection in first two patients in Korea. *J Korean Med Sci* 2020;35(7):e86.

[0004] Liu L, Wang P, Nair MS, et al. Potent neutralizing antibodies against multiple epitopes on SARS-CoV-2 spike. *Nature* 2020;584(7821):450-456.

[0005] Munster JV, Feldmann F, Williamson BN, et al. Respiratory disease in rhesus macaques inoculated with SARS-CoV-2. *Nature* 2020;(May):Accepted preview.

[0006] Murato AE, Fontes-Garfias CR, Ren P, et al. A high-throughput neutralizing antibody assay for COVID-19 diagnosis and vaccine evaluation. *Nat Commun* 2020;11(1):4059.

[0007] Pallesen J, Wang N, Corbett KS, et al. Immunogenicity and structures of a rationally designed prefusion MERS-CoV spike antigen. *Proc Natl Acad Sci USA* 2017;114(35):E7348-57.

[0008] Pardi N, Hogan MJ, Pelic RS, et al. Zika virus protection by a single low-dose nucleoside-modified mRNA vaccination. *Nature* 2017;543(7644):248-51.

[0009] Pardi N, Parkhouse K, Kirkpatrick E, et al. Nucleoside-modified mRNA immunization elicits influenza virus hemagglutinin stalk-specific antibodies. *Nat Commun* 2018;9(1):3361.

[0010] Sahin U, Kariko K, Tureci O. mRNA-based therapeutics - developing a new class of drugs. *Nat Rev Drug Discov* 2014;13(10):759-60.

[0011] Sedic M, Senn J, Lynn A, et al. Safety evaluation of lipid nanoparticle-formulated modified mRNA in Sprague-Dawley rat and cynomolgous monkey. *Vet Pathol* 2018;55(2):341-54.

[0012] Sellers RS, Nelson K, Bindu Bennet B, et al. Scientific and Regulatory Policy Committee Points to Consider: Approaches to the Conduct and Interpretation of Vaccine Safety Studies for Clinical and Anatomic Pathologists. *Toxicol Pathol* 2020;48(2):257.

[0013] Singh DK, SR Ganatra, B Singh et al. SARS-CoV-2 infection leads to acute infection with dynamic cellular and inflammatory flux in the lung that varies across nonhuman primate species. *bioRxiv* 10.1101/2020.06.05.136481 (2020).

[0014] Sliansky JE, Rattis FM, Boyd LF, et al. Enhanced antigen-specific antitumor immunity with altered peptide ligands that stabilize the MHC-peptide-TCR complex. *Immunity* 2000; 13(4):529-38.

[0015] Truong B, Allegri G, Liu X-B, et al. Lipid nanoparticle-targeted mRNA therapy as a treatment for the inherited metabolic liver disorder arginase deficiency. *Proc Natl Acad Sci USA*. 2019;116(42):21150-9.

[0016] US Department of Health and Human Services, Food and Drug Administration, Center for Biologics Evaluation and Research. Development and licensure of vaccines to prevent COVID-19. In: Guidance for industry. Rockville, MD: Food and Drug Administration; 2020.

[0017] World Health Organization. Annex 1. WHO guidelines on nonclinical evaluation of vaccines. In: World Health Organization. WHO technical report series, no. 927. Geneva, Switzerland: World Health Organization; 2005. p. 31-63.

[0018] World Health Organization. Annex 2. Guidelines on the nonclinical evaluation of vaccine adjuvants and adjuvanted vaccines. In: WHO technical report series no. 987. Geneva, Switzerland: World Health Organization; 2014. p. 59-100.

[0019] Wrapp D, Wang N, Corbett KS, et al. Cryo-EM structure of the 2019-nCoV spike in the prefusion conformation. *Science* 2020;367(6483):1260-3.

[0020] Wu Y, Wang F, Shen C, et al. A noncompeting pair of human neutralizing antibodies block COVID-19 virus binding to its receptor ACE2. *Science* 2020; 368(6496):1274-8.

[0021] Xie X, Murato A, Lokugamage KG, et al. An infectious cDNA clone of SARS-CoV-2. *Cell Host Microbe* 2020; 27(5):841-848.e3.

[0001] Xie X, Murato A, Lokugamage KG, et al. An infectious cDNA clone of SARS-CoV-2. *Cell Host Microbe* 2020; 27(5):841-848.e3.

[0002] Yong CY, Ong HK, Yeap SK, et al. Recent advances in the vaccine development against middle east respiratory syndrome-coronavirus. *Front Microbiol* 2019;10:1781.

[0003] Yuan M, Wu NC, Zhu X, et al. A highly conserved cryptic epitope in the receptor-binding domains of SARS-CoV-2 and SARS-CoV. *Science* 2020; 368(6491):630-3.

[0004] Zakhartchouk AN, Sharon C, Sathkumarajah M, et al. Immunogenicity of a receptor-binding domain of SARS coronavirus spike protein in mice: implications for a subunit vaccine. *Vaccine* 2007;25(1):136-43.

[0005] Zhou M, Zhang X, Qu J. Coronavirus disease 2019 (COVID-19): a clinical update. *Front Med* 2020(Apr):1-10.

[0006] Zost Seth J, Gilchuk Pavlo, Chen Rita E, et al. Rapid isolation and profiling of a diverse panel of human monoclonal antibodies targeting the SARS-CoV-2 spike protein *Nature Medicine* 2020;26 9:1422-1427.

[0007] Zou L, Ruan F, Huang M, et al. SARS-CoV-2 viral load in upper respiratory specimens of infected patients. *N Engl J Med* 2020;382(12):03:1177-9.

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 - ▶  [0001] VR-MVR-10081 - Method Validation Report for the Elecsys Anti-SARS-CoV-2 Assay
 - ▶  [0001] VR-MVR-10083 - Validation Report for the SARS-CoV-2 mNeonGreen Virus Microneutralization Assay
 - ▶  [0001] VR-MQR-10211 - Qualification Report for a Single-plex Direct Luminex Assay (dLIA) for Quantitation of IgG Antibodies to SARS-CoV-2 S1 Protein in Human Sera
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 - ▶  [0001] VR-TM-10304 - Test Method for the SARS CoV-2 Nucleocapsid (N) Antigen Detection Assay
 - ▶  [0001] VR-SOP-LC-11120 - Data Review Procedures for Direct Luminex Immunoassays in LIMS v6
 - ▶  [0001] SHI-SOP-10011 - Manual 96-well Neutralization Assay for the Detection of Functional Antibodies to SARS-CoV-2 in Test Serum using Cytation 7 Image Reader
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 - ▲  [0001] VR-MVR-10080 - Report on Method Validation of a Cepheid Xpert Xpress PCR Assay to Detect SARS-CoV-2
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 -  [0001] VR-MVR-10080-ATT01
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 - ▲  [0001] VR-MVR-10081 - Method Validation Report for the Elecsys Anti-SARS-CoV-2 Assay
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 -  [0001] VR-MVR-10081
 -  [0001] VR-MVR-10081-ATT01
 - ▲  [0001] VR-MVR-10083 - Validation Report for the SARS-CoV-2 mNeonGreen Virus Microneutralization Assay
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 -  [0001] VR-MVR-10083
 -  [0001] VR-MVR-10083-ATT01
 -  [0001] VR-MVR-10083-ATT02
 -  [0001] VR-MVR-10083-ATT03
 -  [0001] VR-MVR-10083-ATT04
 - ▲  [0001] VR-MQR-10211 - Qualification Report for a Single-plex Direct Luminex Assay (dLIA) for Quantitation of IgG Antibodies to SARS-CoV-2 S1 Protein in Human Sera
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- ▲ [0001] VR-MQR-10211 - Qualification Report for a Single-plex Direct Luminex Assay (dLIA) for Quantitation of IgG Antibodies to SARS-CoV-2 S1 Protein in Human Sera
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- ▲ [0001] VR-TM-10293 - Single-plex Luminex Assay for Quantitation of IgG Antibodies to SARS-CoV-2 S1 Protein in Human Serum
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- ▲ [0001] VR-TM-10294 - Single-plex Luminex Assay for Quantitation of IgG Antibodies to SARS-CoV-2 RBD Protein in Human Serum
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- ▲ [0001] VR-TM-10298 - Manual 96-well Neutralization Assay for the Detection of Functional Antibodies to SARS-CoV-2 in Test Serum
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- ▲ [0001] VR-TM-10298 - Manual 96-well Neutralization Assay for the Detection of Functional Antibodies to SARS-CoV-2 in Test Serum
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 - [0001] VR-TM-10298
- ▲ [0001] VR-TM-10304 - Test Method for the SARS CoV-2 Nucleocapsid (N) Antigen Detection Assay
 - ▲ Study Report Body Chapter
 - [0001] VR-TM-10304
- ▲ [0001] VR-SOP-LC-11120 - Data Review Procedures for Direct Luminex Immunoassays in LIMS v6
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 - [0001] VR-SOP-LC-11120
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